सत्यमेव जयते

Model Guidelines for the Development of Cold Water Fisheries



Department of Fisheries
Ministry of Fisheries, Animal Husbandry and Dairying
Government of India

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1. Introduction

India's cold-water fisheries represent a distinctive aspect of the country's aquaculture sector, primarily flourishing in the high-altitude areas of the Himalayas and certain regions of the Northeast and peninsular region. These ecosystems create optimal conditions for species such as Rainbow Trout, Brown Trout, Mahseer, and Snow Trout. Despite their potential, cold-water fisheries encounter various challenges, including insufficient infrastructure, a lack of standardized breeding and feeding practices, limited market access, etc. The delicate mountain ecosystems also require environmentally sensitive methods for fish farming.

Sustainability lies at the core of cold-water fisheries management, ensuring that ecological balance is preserved while meeting human needs. Scientific stocking practices, biodiversity conservation, responsible introduction of exotic species, and strong disease-management frameworks are critical to safeguarding fragile mountain ecosystems. Equally important is the meaningful participation of local communities, where stakeholders are actively involved in decision-making and benefit-sharing, fostering a sense of ownership and collective responsibility. The knowledge, stewardship, and co-management roles of these communities remain central to the long-term sustainability and resilience of cold-water fisheries.

Effective implementation of cold-water fisheries guidelines requires strong coordination among departments such as Fisheries, Water Resources, Revenue, Forest, Rural Development, Panchayati Raj, and Local Governance. An integrated, multi-sectoral approach enables better management of water quality, habitat conservation, and socio-economic development. Strengthened inter-departmental synergy and scientific interventions will help unlock the substantial potential of cold-water fisheries in India, leading to higher productivity, sustainable resource use, and improved livelihoods across the hill states.

In this context, the Department of Fisheries, Government of India, has undertaken several initiatives to promote cold-water aquaculture, including support for raceways, hatcheries, Recirculatory Aquaculture Systems (RAS), and modern broodstock management, supply chain infrastructure, and cold chain network. This guideline aims to provide strategic direction to States and Union Territories, enabling them to develop region-specific policies and frameworks, strengthening technical capacities, and ensuring the ecologically responsible expansion of cold-water fisheries. It outlines a structured approach covering species selection, hatchery design, seed production, disease management, infrastructure and technology requirements, and market linkages, while prioritizing sustainability and capacity building. By aligning with national initiatives such as PMMSY and leveraging the technical expertise of institutions like ICAR-CICFR and NFDB, these guidelines aim to unlock the full potential of cold-water aquaculture in India’s upland regions.

2. Cold-water Fisheries in India: Overview

The term "cold-water" describes aquatic ecosystems characterized by low temperatures and high dissolved oxygen levels, creating ideal conditions for specialized fish species such as Rainbow Trout, Mahseer, and Snow Trout.

These ecosystems are primarily located in the Himalayan states and include upland rivers, streams, natural lakes at various altitudes, and reservoirs. They are characterized by features such as fast-flowing waterfalls, cascading rapids, deep pools, and substrates composed of bedrock, boulders, and sand. The diverse range of native and exotic cold-water fish species in these mountainous regions offers significant opportunities for both aquaculture and, to a lesser extent, capture fisheries. Although capture fisheries and carp-based aquaculture contribute modestly to overall freshwater fish production in these areas, their significance remains noteworthy.

Commercial fisheries in cold-water regions primarily rely on stocking lakes and reservoirs, where introducing fry and fingerlings is a crucial strategy for enhancing fish production. However, cultivating indigenous cold-water species has not been economically viable due to their slow growth rate and low yields. Consequently, farming of exotic species like Rainbow Trout has become more prominent. Rainbow Trout, along with Snow Trout and Brown Trout, has adapted well to India's agro-climatic conditions, making it a strong candidate for commercial aquaculture. Additionally, Mahseer, exotic carps, and Indian Major Carps also show promise for cultivation in appropriate cold-water environments.

Significant investments have been made to strengthen cold-water fisheries across the Himalayan and Northeastern states. The Department of Fisheries has approved ₹2,299.56 crore under PMMSY, Blue Revolution, and FIDF to expand infrastructure and enhance productivity. Key outcomes include the construction of over 5,600 raceways, the establishment of 54 trout hatcheries, 5,599 trout-rearing units, 293 cold storage and ice plants, 260 feed mills, and 8,044 transport vehicles. These interventions, together with insurance coverage and livelihood support for millions of fishers, have substantially strengthened the cold-water fisheries value chain and laid a robust foundation for accelerating growth and resilience in India’s upland regions.

To further accelerate growth in cold-water fisheries, India aims to increase production from 1,500 MT in 2019 – 20 to **over 50,000 MT by 2030.** This will be driven by enhancing the productivity of trout raceways from the current 1 tonne per raceway to 2 - 4 tonnes alongside the development of trout and carp hatcheries, construction of ponds and fingerling-rearing units, and expansion of mini and medium-scale RAS facilities. Complementary infrastructure, including cages in cold-water reservoirs, fish transport systems, and value-addition units, will also be strengthened. Parallel efforts to promote polyculture of exotic carps, mahseer, and snow trout in upper hill regions, together with composite carp farming in lower belts, will further diversify and expand overall production potential.

3. Classification of Cold-Water Fish Culture

Cold-water fish culture systems can be broadly classified as follows:

a. <u>Based on the Water Source</u>:

i. Riverine Fish Culture: Utilizes flowing water from rivers and streams.
ii. Reservoir Fish Culture: Conducted in man-made or natural reservoirs.
iii. Lake Fish Culture: Takes place in natural or artificial lakes.
iv. Spring and Hill Stream: Uses perennial springs and hill streams, common in mountainous regions.



Figure 1: Riverine Fish Culture in Sindhu Ghat, Leh, UT, Ladakh

b. <u>Based on Culture System</u>:

i. Extensive Culture: Natural productivity with minimal human intervention.
ii. Semi-Intensive Culture: Combines natural productivity with supplementary feeding and moderate management.
iii. Intensive Culture: Involves high stocking density, regular feeding, and advanced management practices (such as raceways, RAS, etc.)

c. <u>Based on Species Cultured</u>:

i. Indigenous Species Culture: Focuses on native cold-water species such as snow trout, mahseer, minor carps, and other hill stream fishes.
ii. Exotic Species Culture: Involves introduced species like rainbow trout, brown trout, and exotic carps.
iii. Composite culture: Cultivation of multiple compatible species together to maximize resource use.



Figure 2: Rainbow trout farming in Damsna, Kargil, UT, Ladakh

d. Based on Production Objective:

i. Table Fish Production: Aimed at producing fish for human consumption.
ii. Seed Production: Focuses on breeding and rearing fingerlings for stocking / sale.
iii. Ornamental Fish Culture: Rearing of cold-water ornamental species for the aquarium trade.

e. Based on the Culture Environment:

i. Open Water Culture: Fish are cultured in natural open water bodies (rivers, lakes, reservoirs).
ii. Confined Water Culture: Fish are reared in controlled environments such as raceways, tanks, polyhouses, or cages.

f. Based on Management Practices:

i. Monoculture: Rearing a single species in a given water body.
ii. Integrated Fish Farming: Combining fish culture with other agricultural or livestock activities (e.g., fish-cum-poultry, fish-cum-horticulture).

4. Species and Geographic Focus

The geographic distribution of cold-water fish species is intricately connected to altitude, water temperature, and flow characteristics. Rainbow trout and Brown Trout thrive in high-altitude areas with perennial cold streams, which are prevalent in regions like Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh, and Ladakh. Snow Trout and Mahseer, as native species, have a broader distribution throughout the Himalayan belt and can be cultivated in both natural and controlled

environments in states such as Uttarakhand, Sikkim, Arunachal Pradesh, Nagaland, Manipur, Himachal Pradesh, parts of Assam, Meghalaya, and the Jammu region too.

a. Key species include:

i. Trout (Rainbow and Brown)
ii. Mahseer (Golden, Chocolate, and others)
iii. Snow Trout and other indigenous cold-water species

b. Geographic focus areas include:

i. **Himalayan states:** Jammu & Kashmir, Himachal Pradesh, Uttarakhand, and Ladakh, Darjeeling (West Bengal)
ii. **Northeastern states**: Arunachal Pradesh, Sikkim, Meghalaya, Nagaland, Manipur, and Assam.
iii. **Other states:** West Bengal, Tamil Nadu, Kerala, and Karnataka



Figure 3: Geographical regions for Coldwater Fish Farming

c. Species Profiles:

i. ***Rainbow Trout (Oncorhynchus mykiss):*** An exotic species introduced from Europe, Rainbow Trout is highly valued for its fast growth, palatability, and adaptability to Indian cold-water conditions. It thrives well in water temperatures between 8°C and 18°C and is widely cultured in Himachal Pradesh, Jammu & Kashmir, and Uttarakhand.



Figure 4: Rainbow trout

ii. ***Brown Trout (Salmo trutta):*** Another exotic species, Brown Trout is known for its resilience and is often used for recreational and sport fishing. It prefers cooler waters and is found in high-altitude streams and rivers of Jammu & Kashmir, Uttarakhand, Arunachal Pradesh, Sikkim, and Himachal Pradesh.



Figure 5: Brown trout

iii. ***Snow Trout (Schizothorax spp.):*** An indigenous species, Snow Trout is well adapted to the cold, fast-flowing rivers of the Himalayas. It is important for both ecological balance and local consumption, with potential for semi-commercial culture in regions like the Indian Himalayan Region (IHR).



Figure 6: Snow trout (Schizothorax spp)

iv. ***Mahseer (Tor spp.):*** A culturally and ecologically significant indigenous species, Mahseer is found in the rivers of Uttarakhand, Himachal Pradesh, and Northeast India. It is a slow-growing species but holds promise for conservation, aquaculture, and eco-tourism.



Figure 7: Golden Mahseer

d. Promotion of Indigenous Species: To balance ecological sustainability with economic development, greater emphasis is placed on promoting indigenous cold-water species such as Mahseer and Snow Trout alongside exotic varieties. These native species hold significant potential for conservation, eco-tourism, and niche global markets. Leveraging AI-based habitat mapping and geospatial technologies, in collaboration with institutions like ISRO and the Department of Space, can enhance targeted conservation efforts and support strategic planning for indigenous species management.

The development of cold-water aquaculture should prioritize ecological sustainability. The introduction of exotic species shall be carefully regulated to avoid negative effects on native biodiversity. Key practices such as habitat conservation, water quality monitoring, and responsible stocking are crucial for preserving the ecological integrity of mountain aquatic ecosystems. Furthermore, combining traditional knowledge with scientific methods can improve the resilience and acceptance of cold-water fisheries within local communities.

5. Objective of the Model Guidelines

The broad objectives of the guidelines are to:

i. Enhance fish production and productivity through scientific and sustainable practices.
ii. Enable long-term ecological balance of cold-water fisheries while meeting the needs of present and future generations.
iii. Guide on strategies to protect and enhance the diversity of aquatic species and habitats in cold water ecosystems.
iv. Strengthen institutional frameworks and governance mechanisms for cold-water fisheries.
v. Generate rural employment and promote entrepreneurship, particularly among women and youth.

vi. Introduce effective protocols for disease prevention, monitoring, and health management of aquatic organisms.
vii. Facilitate market linkages and value-chain integration for cold-water fisheries.
viii. Encourage adoption of modern technologies, including Remote sensing, GIS mapping, IoT-based monitoring etc.
ix. Provide strategic recommendations to States and Union Territories for formulating tailored policies and guidelines suited to the local conditions.
x. Ensure that the sustainable development of cold-water fisheries contributes to improved livelihoods and economic well-being of the fishing communities.

6. Key Considerations for Cold-Water Aquaculture Development

6.1. Assessment of Site and Water Quality

Site selection and water quality are vital factors influencing the success of cold-water aquaculture. Due to the sensitivity of cold-water species to environmental conditions, it is crucial to thoroughly assess site characteristics and continuously monitor water quality parameters to ensure sustainable and productive fish farming.

a. Site Selection Guidelines:

Ideal site characteristics for cold-water fisheries include:

i. **Altitude**: Preferably above 1,500 meters for Trout culture; lower altitudes may be suitable for Mahseer.
ii. **Water Source**: Perennial, clean, and cold-water sources such as snow-fed streams, springs, or rivers.
iii. **Flow Rate**: Adequate flow to ensure oxygenation and waste removal; minimum 2-3 Liters/sec for small units.
iv. **Accessibility:** Proximity to roads and markets for ease of transport and logistics.
v. **Land Topography:** Gently sloping or terraced land to facilitate gravity-fed water systems.

b. Water Quality Parameters:

Key water quality parameters for cold-water fish species include,

i. **Temperature:** Optimal range is 8°C to 18°C for Trout; Mahseer may thrive well in temperatures of 20-25°C.
ii. **Dissolved Oxygen (DO):** Minimum 6 mg/L; higher levels preferred for Trout.
iii. **pH Value:** Ideal range is 6.5 to 8.0.
iv. **Turbidity:** Preferably low; high turbidity affects gill function and feeding.
v. **Ammonia and Nitrite Content:** Preferably negligible; toxic to fish even at low concentrations.

6.2. Key Infrastructure and Technology Requirements

Infrastructure and technology form the backbone of successful cold-water aquaculture, enabling efficient production, health management, and market connectivity in challenging high-altitude environments. Given the sensitivity of cold-water species like Rainbow Trout and Mahseer to environmental conditions, specialized infrastructure is essential to maintain optimal water quality, temperature, and oxygen levels throughout the production cycle.

a. Key infrastructure requirements may include:

i. **Hatcheries:** Equipped with flow-through systems, temperature control, and oxygenation units to support breeding and seed production.

ii. **Raceways and Ponds:** Designed to facilitate water circulation, waste removal, and ease of management.

iii. **Feed Mills and Storage Units:** For producing and storing high-protein, species-specific feed formulations.

iv. **Cold Chain Logistics:** Including insulated transport vehicles and storage facilities to maintain fish quality during distribution.

v. **Water Quality Monitoring Stations:** To continuously track parameters like temperature, pH, DO, ammonia, and turbidity.



Figure 8: ICAR-CICFR established Trout Hatchery at Dechenthang Mechuka

b. Technological Interventions:

i. **Automated Feeders:** Improve feed efficiency and reduce labour costs.

ii. **Recirculatory Aquaculture Systems (RAS):** Recirculatory Aquaculture Systems (RAS), already piloted by startups, offer a scalable and water-efficient model for sustainable production.

iii. **Sensors and IoT Devices**: Enable real-time monitoring of water quality and fish health.

iv. **GIS Mapping, Digital Dashboards and Platforms:** Assist in site selection, resource planning, environmental monitoring, and disease forecasting. A national cold water aquaculture database covering seed supply, farm performance, and health status may be created to strengthen planning and response mechanisms. Digital Platforms may be developed for farm management, traceability, and market access.

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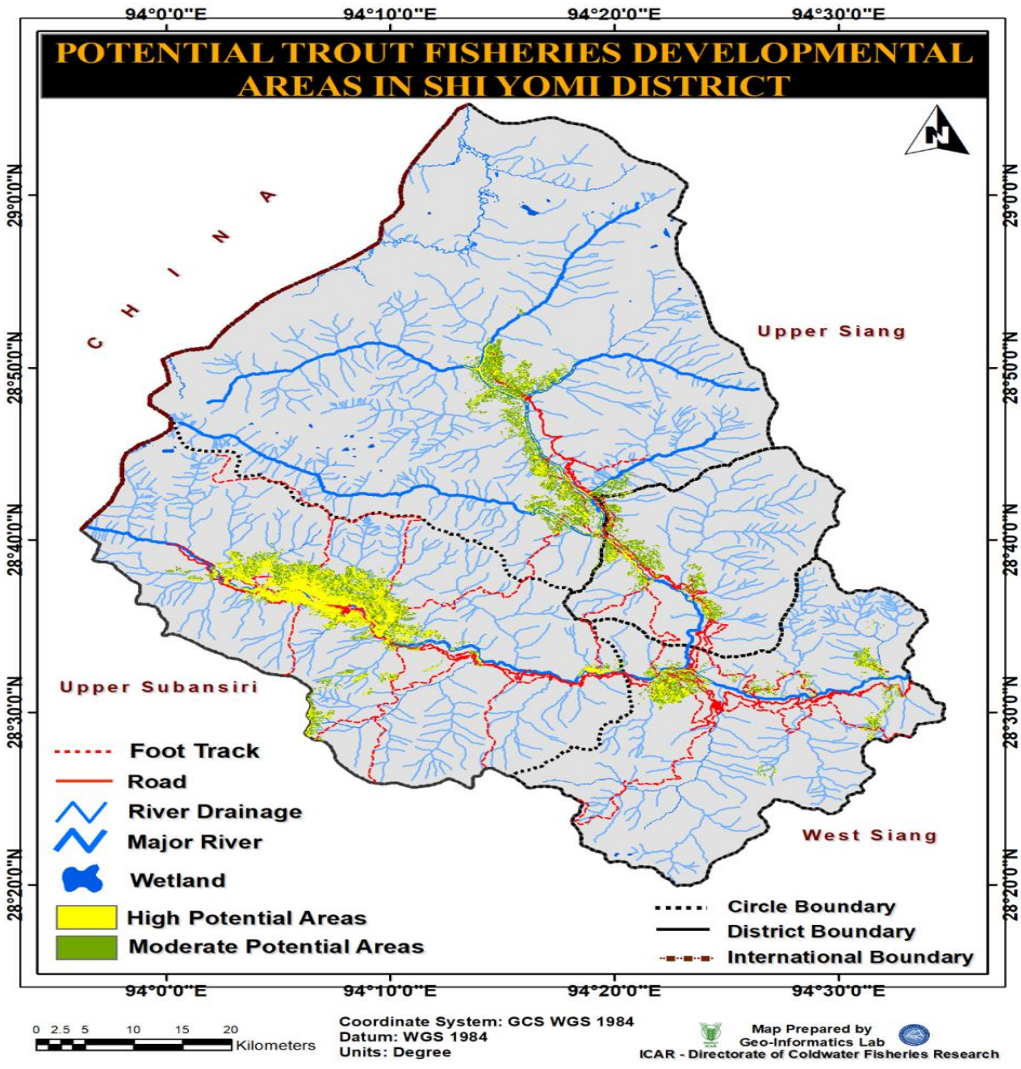


Figure 9: GIS Mapping and Remote Sensing to assist site selection

Table 1: Following Technologies certified by ICAR-FS-CICFR (earlier DCFR) may also be referred on the basis of requirements, in consultation with the institute:

Reference	Name of the Technology certified by ICAR-FS-DFCR
ICAR-FS-DCFR- Technology-2023- 059	Captive maturation and multiple breeding of golden mahseer: *Technology*
ICAR-FS-DCFR-Technology-2023- 060	Breeding and seed Production technology of Suckerhead, *Garra gotyla*: *Technology*
ICAR-FS-DCFR-Technology-2023- 061	Technology for sterile triploid rainbow trout (*Oncorhynchus mykiss*) production: Technology
ICAR-FS-DCFR-Product-2023-062	High Performance Rainbow trout starter feed DCFR-PerforMAX-S: Product
ICAR-FS-DCFR-Technology-2023- 005	High Performance Rainbow trout Grower feed DCFR-PerforMAX-G: Product
ICAR-FS-DCFR-Technology-2023- 063	DCFR Aqua FSD, A novel anaesthetic formulation of fishes: Product
ICAR-FS-DCFR-Product-2023-064	Cell line from snow trout muscle (SRM-1): Product
ICAR-FS-DCFR-Product-2023-065	Cell line from rainbow trout heart (RBT- H): Product
ICAR-FS-DCFR-Technology-2023- 066	Breeding protocol of *Schizothorax progastus:* Technology
ICAR-FS-DCFR-Technology-2023- 067	Breeding protocol of *Schizothorax plagiostomus:* Technology
ICAR-FS-DCFR- Technology-2024-039	Recirculating Aquaculture System based Intensive Farming of Rainbow Trout.
ICAR-FS-DCFR- Product-2024-041	DCFR-MAHA-Brood: Mahseer brooder feed for enhanced reproductive performance.
ICAR-FS-DCFR- Product-2024-040	DCFR Lg Kit for identification of bacterial pathogen *Lactococcus garvieae.*
ICAR-FS-DCFR- Protocol-2024-038	Live transportation protocol for different stages of Rainbow trout (DCFR-ShipSafe).
ICAR-FS-CICFR-Technology-2025-044	Recirculating aquaculture system-based hatchery and nursery technology for rainbow trout
ICAR-FS-CICFR-Product-2025-044	Visual nano-biosensor for identification of oomycete pathogen, *Saprolegnia parasitica*
ICAR-FS-CICFR-Product-2025-044	Ovaqua: A hormonal formulation for induced breeding in fish
ICAR-FS-CICFR-Product-2025-044	High performance rainbow trout brooder feed, DCFR-PerforMAX-B

As per ICAR-Directorate of Coldwater Fisheries Research (DCFR), Bhimtal (Annual Report 2023- 2024)

6.3. Considerations for designing Hatcheries and Seed Production

a. **Hatchery Design**: An effective hatchery features flow-through systems, temperature control, and oxygenation mechanisms to replicate natural cold-water conditions.

b. **Seed Production**:
   i. Requires careful selection of broodstock, controlled spawning, incubation, and larval rearing under optimal water quality conditions.
   ii. It is recommended to ensure genetic diversity, disease-free stock, and high survival rates during the early life stages to maintain productivity and sustainability.
   iii. Developing region-specific hatchery models and standard operating procedures is recommended for increasing seed availability and supporting both commercial and conservation-focused cold-water fish farming.
   iv. Ensuring a steady supply of healthy juvenile fish is essential for maintaining the genetic quality and survival rate.



Figure 10: Transport of eyed ova

6.4. Disease Prevention and Health Management

Effective disease prevention and health management are essential for maintaining the productivity and sustainability of cold-water fisheries, especially given the sensitivity of species like Rainbow Trout, Brown Trout, and Mahseer to environmental stressors. Coldwater fish are particularly vulnerable to bacterial, fungal, and parasitic infections, which can spread rapidly in high-density culture systems if not properly managed.

a. Preventive Measures:

i. **Biosecurity Protocols:** Implement strict biosecurity measures, including controlled access to farms, disinfection of equipment, and quarantine of new stock.

ii. **Water Quality Monitoring:** Maintain optimal water parameters such as temperature, dissolved oxygen, pH, and turbidity to reduce stress and susceptibility to disease.

iii. **Stocking Density Management**: Avoid overcrowding to minimize stress and reduce the risk of disease outbreaks.

iv. **Vaccination and Probiotics:** Use available vaccines and probiotics to enhance immunity and reduce reliance on antibiotics.

v. **Sanitation and Waste Management:** Regular cleaning of raceways, ponds, and cages, along with proper disposal of organic waste, helps prevent pathogen buildup.

b. Health Surveillance and Response:

i. **Routine Health Checks:** Regular visual inspections and sampling to detect early signs of disease such as abnormal swimming, discoloration, or lesions.

ii. **Diagnostic Support:** Collaborate with veterinary labs and fisheries institutions for accurate diagnosis and treatment recommendations.

iii. **Treatment Protocols:** Use approved therapeutics and follow dosage guidelines strictly to avoid resistance and environmental contamination.

iv. **Record Keeping:** Maintain detailed logs of health status, treatments administered, and outcomes to inform future management decisions.

v. **Use of NASPAD - Mobile Application**: The NASPAD mobile application "*ReportFishDisease*" may be systematically integrated into cold-water aquaculture practices to enable real-time monitoring, early detection, and effective management of fish diseases. By leveraging its diagnostic tools and data-sharing capabilities, farmers and field officers can promptly report outbreaks, access expert guidance, and implement timely interventions. Collaboration with research institutions and fisheries departments will

further enhance the app's utility as a digital surveillance and decision-support system.

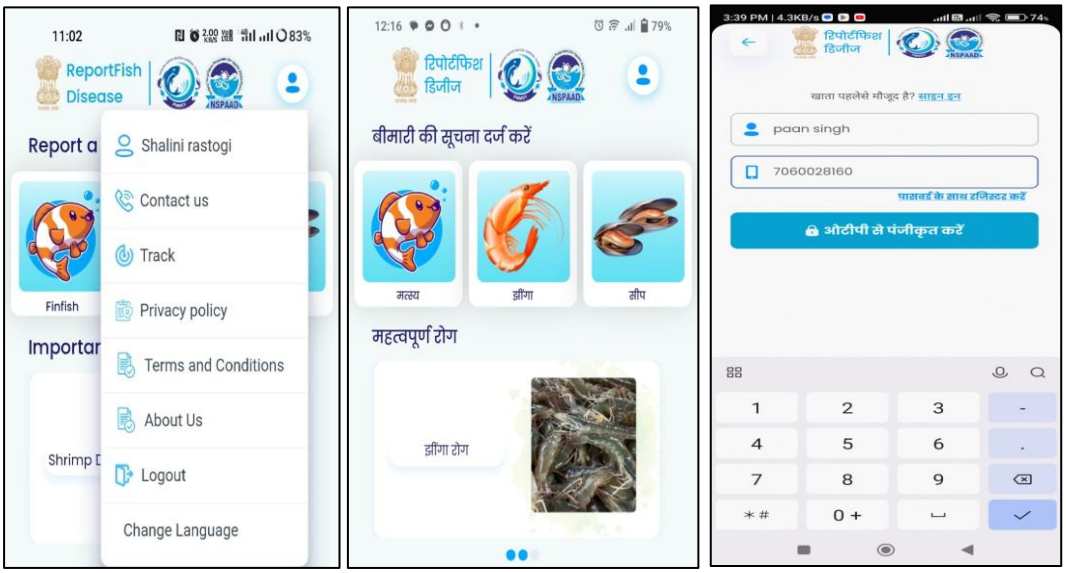


Figure 11: "ReportFishDisease" Application

6.5. Market Linkages and Value Chain Development

The following measures may be taken for market linkages and value chain development:

i. Development and upgradation of facilities for cold storage, transportation, and processing to maintain product quality and reduce post-harvest losses.
ii. Introduction of region-specific branding and certification schemes to enhance consumer trust and market value of cold-water fish products.
iii. Establishment of Fish Farmer Producer Organizations (FFPOs) to improve collective bargaining and streamline supply chains.
iv. Promotion of processed and packaged cold-water fish for product diversification (e.g., smoked trout, ready-to-cook fish) to increase profitability and shelf life.
v. Integration of digital marketing platforms and mobile apps to expand market reach and enable traceability.
vi. Establishment of mechanisms for collecting and analysing market data to guide production planning, pricing strategies, and demand forecasting.
vii. Partnerships with hotels, restaurants, and tourism operators to position cold-water fish as a premium product.

6.6. Cold Water Fisheries Development and Livelihood Diversification

The following measures could be considered for Cold Water Fisheries Development and Livelihood Diversification:

i. Promotion of both indigenous and exotic cold-water fish species suited to local ecological conditions.

ii. Promotion of integrated fish farming with agriculture, horticulture, and livestock to diversify income sources and optimize resource use.
iii. Provision of access to credit, insurance, and subsidies to fishers and entrepreneurs for business development and risk mitigation.
iv. Support for processing, packaging, and branding of cold-water fish products to improve marketability and income.
v. Formation of Fish Farmer Producer Organizations (FFPOs) and cooperatives to strengthen market access and bargaining power.
vi. Design of targeted programs to engage women and youth in fisheries-related activities, including hatchery operations, feed preparation, ornamental fish culture, and marketing.
vii. Development of recreational fishing zones and eco-tourism initiatives around cold-water bodies to generate alternative livelihoods.

6.7. Capacity Building and Training

Training programs may be designed to cover the entire value chain of cold-water aquaculture, including:

i. Site selection,
ii. Species-specific culture techniques,
iii. Water quality management,
iv. Seed production protocols,
v. Feed formulation and preparation,
vi. Health surveillance, and
vii. Post-harvest handling and processing

These programs may be delivered in local languages and tailored to regional contexts to ensure accessibility and relevance. Hands-on demonstrations, exposure visits, and farmer field schools can be effective methods for knowledge transfer.

Establishment of *"Model Coldwater Aquaculture Clusters"* and demonstration farms in strategic locations is recommended. These clusters may showcase raceway trout farming, mahseer conservation, and snow trout culture while serving as hubs for farmer field schools, exposure visits, and hands-on training. Such model centres may also function as innovation platforms for testing improved feed, seed, and management practices.

Institutions such as ICAR, NFDB, and State Fisheries Departments shall take the lead in developing standardized training modules and certification programs. Additionally, partnerships with universities, NGOs, and private sector players may help scale outreach and innovation. By investing in capacity building, states / UTs may empower local communities, improve farm productivity, reduce risks, and foster entrepreneurship in cold-water fisheries. Strengthening international collaboration is vital for advancing cold-water fisheries. Strategic technology transfer and formal MoUs with countries like **Iceland, Denmark, and Norway** may support genetic improvement,

modern trout hatchery systems, broodstock improvement, disease diagnostics & aquatic health management, and value-added product development, post-harvest quality assurance, etc.

7. Governance Framework

7.1. Conservation and Biodiversity Sustainability

The following protective measures may be adopted for the conservation and sustainability of cold-water fisheries in India:

i. Implementation of seasonal fishing bans, size limits, and gear restrictions to prevent overfishing and protect breeding populations.
ii. Promote the conservation of indigenous and endangered cold-water fish species such as mahseer and snow trout through captive breeding and restocking initiatives.
iii. Engagement of local communities in conservation efforts through awareness programs, co-management models, and benefit-sharing mechanisms.
iv. Integration of climate-resilient practices such as habitat buffering, flow regulation, and adaptive stocking strategies to mitigate climate change impacts.
v. Designation and enforcement of eco-sensitive zones around critical cold-water fish habitats to restrict harmful activities.

7.2. Role & Responsibility of State/ UTs (Fisheries Department)

The responsibility of the State/UT Fisheries Department for enhancing cold-water fish production and productivity:

a. Develop and adopt state/UT-specific policies and action plans for cold water fisheries, aligned with national guidelines, and ensure the effective implementation of these policies at district and local levels.

b. Constitute Multi-Stakeholder Advisory Committees at state, district, and local levels, including representatives from relevant departments, fisheries research institutions, and community organizations.

c. Develop robust systems for monitoring progress, evaluating outcomes, and maintaining accountability. This includes establishing comprehensive databases that capture key metrics such as fish production, resource health, and socio-economic indicators. Additionally, regular reporting mechanisms may be instituted across local, district, and state levels to facilitate transparency, informed decision-making, and timely interventions.

d. Enforce fisheries regulations, licensing, and compliance with environmental and quality standards.

e. Promote public-private partnerships and collaboration with NGOs, cooperatives, and private sector entities.

f. Organize training, workshops, and exposure visits for department staff, fishers, and other stakeholders on scientific and sustainable practices and facilitate knowledge sharing and dissemination of best practices.

g. Establish accessible and transparent mechanisms for grievance redressal and conflict resolution among stakeholders. These systems ensure fair and timely resolution of disputes related to resource use, access, and management.

h. For the management of the cold-water fisheries development, a "Coordination Committee" may be formed at the District level and the committee may be led by officer from State/UT Fisheries Department, consisting of the representative from line Department such as Deputy Directors/ Assistant Engineers of (i) Water Resource Department, (ii) Revenue Department, (iii) Forest Department, (iv) Rural Development, and (v) Panchayati Raj. The meeting of the above Committee may be held in every three months or as required.

Functions of the Coordinating Committee:

(I) To recommend a no-objection certificate/ transfer of fisheries rights.
(II) Take necessary initiatives to remove the inter-departmental bottlenecks in the implementation of the fisheries development schemes.
(III) Mark the suitable sites for the construction of cold-water fisheries infrastructure.
(IV) Integrate cold-water fisheries initiatives with other state and central government schemes for rural development, livelihoods, and environmental conservation.
(V) Ensure active participation of local communities, fishers, women, and youth in planning, implementation, and monitoring.
(VI) Oversee sustainable utilization of water bodies, regulate stocking and fishing, and implement habitat conservation measures.

7.3. Evaluation of Cold-water Fisheries Development Projects

i. The assessment of fisheries resources and the development of cold-water fisheries in the region may be conducted by the State/UT Fisheries Department through Fisheries Research Institutes at appropriate intervals.

ii. State/UT Fisheries Department may consult with such Fisheries Research Institutes/Fisheries Universities/Colleges and prepare a proposal with details (assessment fee, time limit for fee payment, points of investigation etc.) and list

them after departmental approval so that initiatives can be taken for assessment on time.

iii. Based on the assessment report, an Action Plan may be prepared for support under the Central and State Fisheries Development schemes and programmes.

7.4. Implementation and Institutional Mechanisms

A dedicated Cold-water Fisheries Division may be constituted with inter-governmental coordination and collaboration with NFDB/ICAR institutes or fisheries universities for effective management and implementation of the above instructions. The States / UTs shall endeavour to adhere to the above-said instructions.

8. Model Technical Specifications for Raceways & RAS

8.1. Pre-stock management

i. The seed rearing tanks may be cleaned before the fry/fingerlings arrive. Once the preparation of the tanks is over, the water supply may be checked.

ii. Trout is sensitive to changes in water temperature, especially when the temperature shifts from cold to warm. The smaller the fish, the more sensitive they are and susceptible to thermal shock. Therefore, it is important to raise or lower the temperature of the transport water slowly, in increments of 0.5 °C/min, to ensure safety.

iii. The optimal velocity of water in the raceway is 2–3 cm/sec (1.2–1.8 m/min) for smaller fish and 4–10 cm/sec (2.4–6 m/min) for larger ones. Minimum water flow of more than 500 litres per minute is suggested for growing one tonne of rainbow trout production.

8.2. Post-stock management

The physico-chemical parameters and hygiene lead to the successful culture of Trout fish:

i. **Optimum temperature**: 8°C to 18°C.

ii. **Dissolved oxygen**: More than 6 mg/L or above 80% DO saturation

iii. **pH Value**: A neutral or slightly alkaline pH is best for the trout. An optimum pH level of 7-8 is ideal for growth.

iv. **Turbidity**: Clear water without any contamination is required for culture. However, the turbidity should not be more than 25 cm on the Secchi disc.

v. **Hygiene**: Cleanliness is a very important factor in Trout farming. The fish may be cleaned and disinfected either with 10% formalin or 4 mg/L $KMnO_4$ solution periodically through dip treatment. The infected fish may be immediately removed from the tank, and due care should be taken to consult a fishery expert regarding the disease, if any.

8.3. Model Technical Specifications for Trout Culture

The following table presents the model technical specifications for trout fish culture:

Table 2: Model specifications for trout fish culture

Sl. No.	Details	Model Specifications
1.	Name of Species	Rainbow Trout (*Oncorhynchus mykiss*)
2.	Tank Size	17m x 2m x 1.5 m (i.e. 51 m^3)
3.	Effective water depth	1.3 m
4.	Number of Tanks	4 No.
5.	Effective water volume	44.2 m^3/Tank
6.	Stocking size	10 g
7.	Stocking density	100 No./ m^3
8.	Stocking no.	5000 No. / Tank
9.	Ideal survival rate	80%
10.	FCR (for grow out)	1.1 -1.5
11.	Culture period/crop duration	7 -10 Months
12.	Cost of seed (fingerling)	Rs. 10 per Seed
13.	Cost of Feed	Rs. 110 to 150 per kg
14.	Total feed required	6 – 8.4 MT
15.	Size at the time of Harvest	250 -350
16.	Expected Total Biomass	4-5.6 MT
17.	Sale Pricing	Rs. 500-600 per kg

8.4. Indicative Layouts

A. Indicative Layout of Re-circulatory Aquaculture System (RAS):

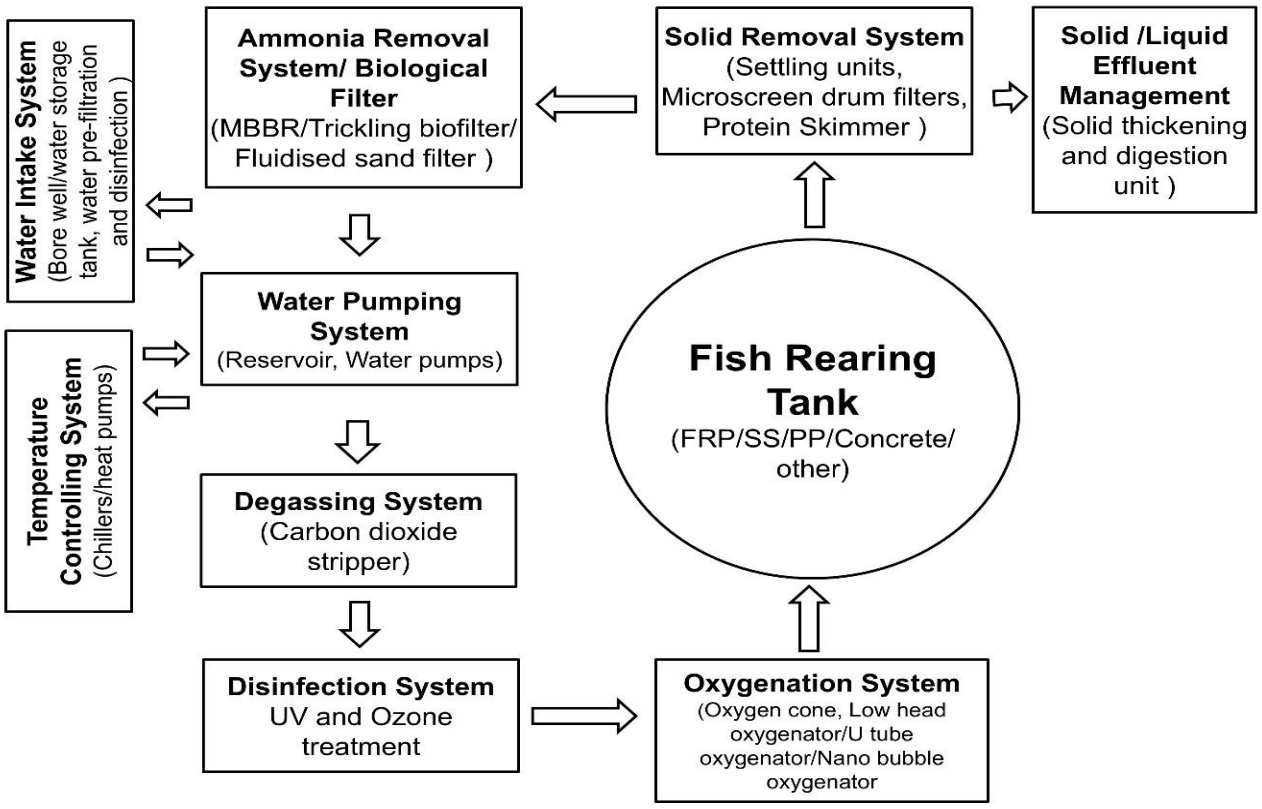


Figure 12: Indicative process flow - recirculating aquaculture system (RAS): Rainbow trout farming

B. Indicative Layout of Raceway:

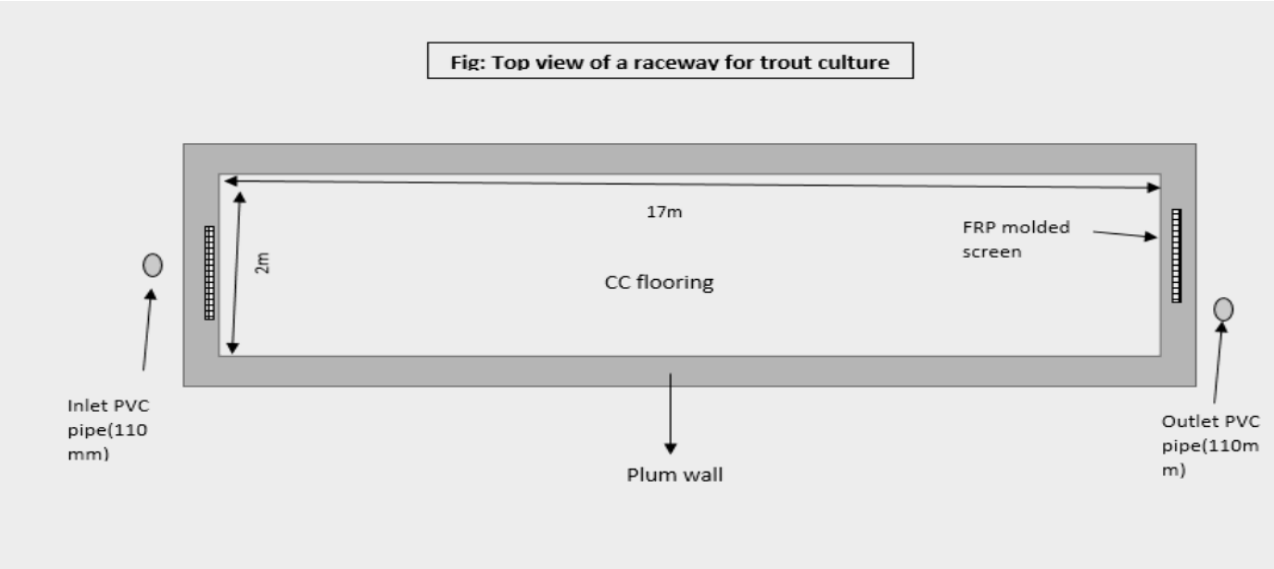


Figure 13: Indicative Layout of Raceway

8.5. Components

a. Indicative Components of Re-circulatory Aquaculture System (RAS):

i. Insulated shed/ Building
ii. Store cum office for feed and accessories
iii. Pump house
iv. Nursery tanks: FRP/PP tanks with inlet and outlet
v. Grow out tanks: Circular cement tanks/ FRP/PP tanks, including inlet, outlet
vi. Settling tanks for sludge, effluent treatment system
vii. Water Storage (sump) tanks and pre-filtration systems
viii. Overhead tanks
ix. Mechanical filters/microscreen filters,
x. Pumps and motors
xi. Power generator
xii. Sludge collector, settable/ dissolved solid collectors
xiii. Bio-filter, and protein skimmer, ozone generator, UV units
xiv. Electrification
xv. Automatic feeder (wherever required)
xvi. Aeration system (air/ oxygen), Carbon dioxide trapper system (degasser),
xvii. Depuration tanks and system
xviii. Temperature controlling system
xix. Water testing kits, water quality monitoring system
xx. Water supply system, bore well etc (wherever required)
xxi. Inputs such as Seed, Feed, additives and supplements, electricity/ Diesel, manpower etc.



Figure 14: Recirculatory Aquaculture System developed at ICAR-CICFR

b. Indicative Components of Raceways:

i. Cemented ponds/ raceways (preferably rectangular),
ii. Inlet and outlet for overflow of water fixed with wire mesh screws, to prevent the escape of stocked species.
iii. Drainage pipe at the bottom of the raceway to facilitate the harvesting as well as the cleaning of the tank from time to time,
iv. Water supply in raceways should be through a filter bed/sedimentation tank, etc.

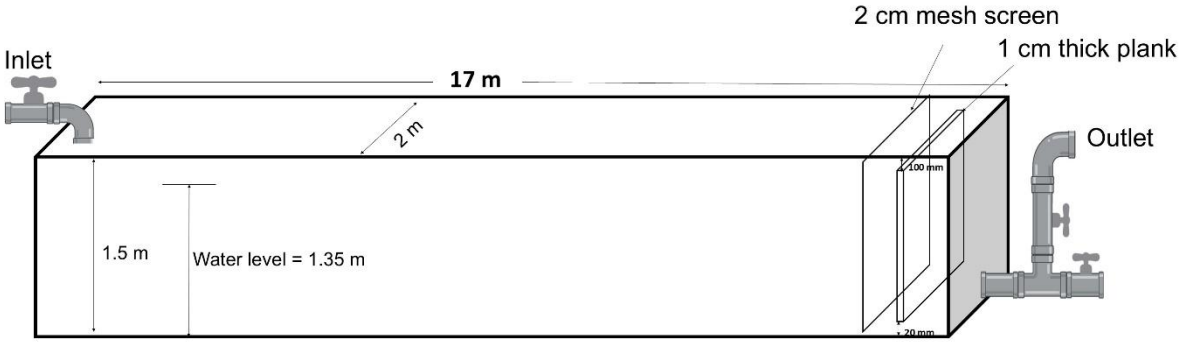


Figure 15: Rainbow trout raceway in Mechuka, Shi-Yomi, Arunachal Pradesh

9. Model Flow-through Raceway Design with Indicate Cost Estimate

A standard flow-through raceway design for the culture of rainbow trout may have dimensions of 17 m length, 2 m width, and 1.5 m total depth, with an operational water depth of 1.35 m, providing an effective water volume of approximately 45 m^3. The raceway is constructed with a longitudinal slope of 0.5–1% (approximately 0.3°–0.6°) towards the outlet to facilitate efficient water flow and waste removal. For a 17 m raceway, this corresponds to a vertical drop of about 15–20 cm from inlet to outlet, ensuring continuous flushing of faecal waste. A 1.5 cm mesh screen (Frame dimension: height 1.5 m and width 2 m) should be provided across the raceway to prevent fish escape until the trout reach about 50 g body weight, after which the net may be removed. A 1 cm thick plank structure (height 1.2 m and width 2 m) with a 20 mm gap from the raceway bottom may be installed to facilitate suction and easy flushing of faecal solids. The inlet pipe size may range from 4 to 6 inches, while the outlet should be a 6-inch pipe fitted with dual gate valves, as shown in the figure, allowing one valve for continuous water flow and the other for periodic cleaning of the raceway when required. However, in hilly areas, it may not always be possible to construct raceways exactly according to the standard design, and the layout may vary depending on site conditions. The site selection should be carried out carefully,

ensuring that the raceway location is at least 20–30 m away from the river, away from potential landslide zones, and accessible by proper road connectivity.



Length: 17 m; Width: 2 m; Total depth: 1.5 m; Operational water depth: 1.35 m; Total water volume: 45 m³; Longitudinal slope: 0.5–1% (0.3°–0.6°) toward the outlet, corresponding to approximately 15–20 cm drop over 17 m length.

Figure 16: An indicative flow-through raceway design for rainbow trout farming

Water flow requirement:

For rainbow trout farming in flow-through raceway systems, stocking density or standing biomass should be determined based on the available water flow at the site. Although rainbow trout can survive in temperatures ranging from 1–22 °C, the optimal growth range is 8–18 °C, with best growth generally observed between 12 and 17 °C. In flow-through raceway systems, water flow is essential to supply dissolved oxygen and flush out ammonia, carbon dioxide, and faecal solids, as fish are maintained at much higher densities than in pond culture. Based on the oxygen requirements of rainbow trout, approximately 500 - 600 litres per minute of water is required to maintain or grow 1,000 kg of rainbow trout biomass at a temperature range of 15–18 °C. Water flow requirements increase as fish grow in body weight or total biomass. The table below presents stage-wise water requirements for a single standard raceway.

Table 3: Average water flow required for sustaining given standing biomass of rainbow trout in flow-through raceway under (12- 18 °C), (LPM = Liter per minute)

Fish Weight (g)	Fish Numbers/ raceway	Biomass (kg/raceway)	Water flow required (LPM/ raceway)
10	5000	50	38
50	4800	240	175
100	4700	470	374
200	4600	920	613
300	4500	1350	875

Table 4: Flow-through raceway dimensions, key production parameters, and cost for as rainbow trout farming

Sl. No.	Details	Model Specifications
1.	Name of Species	Rainbow Trout (*Oncorhynchus mykiss*)
2.	Raceway dimension	17m x 2m x 1.5 m
3.	Water depth	1.35 m
5.	Water volume	45 m^3 per raceway

6.	Stocking size	10 g
8.	Stocking no.	5000 No. per raceway
9.	Survival rate	90 %
10.	FCR (for grow out)	1.1
11.	Culture period/crop duration	8 -12 Months
12.	Cost of seed (fingerling)	Rs. 10 per Seed
13.	Cost of Feed	Rs. 150 per kg
14.	Total feed required	1.48 tonnes
15.	Size at the time of Harvest	300 g
16.	Expected Total Biomass	1.35 tonnes
17.	Sale Pricing	Rs. 600 per kg

Table 5: Indicative Economics of Rainbow Trout Farming in Flow-Through Raceways in Hill Regions (Standard Raceway)

Details	Est. Cost (INR)
Capital cost	
Raceway construction (dimension 17 m x 2 m x 1.5 m)	400000
Plumbing/water intake system, inlets/outlet systems	50000
Other cost (Weighing balance, packaging equipment)	50000
Total capital cost	**500000**
Operating cost (per raceway/ per crop)	
Seed (5000 per raceway) @ 10 Rs per fish	50000
Feeding Cost (FCR 1.1; @ 150 Rs. per kg and 90% survival rate)	222750
Other miscellaneous costs (nets, medicines, disinfectants, packing materials)	25000
Labour cost	24000
Total Operating Cost	**321750**
Total Project Cost	**821750**
Revenue	
Fish Production per Raceway: 1350 kg in 12 months	
Revenue from fish sales (@ 600 per kg)	810000
Net profit (Operating cost basis)	**488250**
Net profit (on total cost basis, including interest)	**431250**
Avg. Benefit-Cost Ratio (BCR):	1.9
Payback period:	1.2 year

Note: Calculations are based on a 14% interest rate and a 10-year life for the raceway.

10. Government-Sponsored Fish Cultivation Initiatives

Government-sponsored schemes play a pivotal role in supporting fish farmers through financial assistance, the adoption of modern technology, market access, and risk-mitigation measures. These programmes collectively aim to enhance production, reduce operational costs, strengthen value-chain infrastructure, and improve the socio-economic well-being of fishers across the country, including those engaged in cold-water aquaculture.

10.1. Benefits to Farmers

Government initiatives provide a wide range of benefits, including:

a) **Financial Assistance** in the form of subsidies and low-interest credit reduces capital and operational costs, enabling farmers to invest in essential infrastructure such as hatcheries, raceways, ponds, and cold-chain facilities.

b) **Modern Technology Adoption,** including Recirculatory Aquaculture Systems (RAS), biofloc, improved feed technologies, and scientific seed production enhance productivity and ensures efficient use of resources through technology-enabled farming.

c) **Market Access and value-chain** strengthening through investments in cold storage, ice plants, processing units, and transport vehicles improves product quality and price realization, complemented by support for branding, certification, and value-added product development.

d) **Skill development and capacity building initiatives** further equip farmers with skills in feed management, water-quality monitoring, disease control, and post-harvest handling.

e) **Risk management** through credit facilities, insurance schemes, and working-capital support helps mitigate financial vulnerability and ensure continuity across production cycles.

10.2. Significant fisheries initiatives/schemes for farmers

10.2.1. Pradhan Mantri Matsya Sampada Yojana (PMMSY)

a) Objective: Enhance fisheries production and productivity, double farmer incomes, and promote modern technologies.
b) Support: Seed, Feed, Production technologies, RAS and biofloc systems, trout and carp hatcheries, raceways, pond development, Cold-chain infrastructure, processing units, and value-chain interventions.
c) Benefits: Increased production, cost reduction, job creation, and strengthened cold-water aquaculture systems.

10.2.2. Fisheries and Aquaculture Infrastructure Development Fund (FIDF)

a) Objective: Creation of large-scale, long-term fisheries infrastructure.
b) Support: Fishing harbours, cold storages, processing units, feed mills, hatcheries, and modern aquaculture systems.
c) Benefits: Reduction in capital burden, improved infrastructure, and enhanced sustainability.

10.2.3. Fisheries Kisan Credit Card (KCC)

a) Objective: Provide timely and affordable working capital to fish farmers.
b) Support: Credit for seed, feed, equipment, and operational requirements.
c) Benefits: Lower financial risk, smoother production cycles, and greater liquidity for farmers.

10.2.4. National Fisheries Development Programme (NFDP)

a) Objective: Strengthen fisheries infrastructure, including landing centres and harbours.
b) Support: Export facilitation, smart harbours, marketing infrastructure, and capacity building.
c) Benefits: Better market prices, improved export potential, and enhanced sectoral competitiveness.

10.2.5. MPEDA Schemes (Marine Products Export Development Authority)

a) Objective: Support export-oriented aquaculture and value-added products.
b) Support: Biosecurity measures, technical innovation, disease-diagnostic facilities, and training.
c) Benefits: Enhanced productivity, improved quality standards, and increased export readiness.

10.2.6. National Fisheries Development Fund (NFD Fund)

a) Objective: Conservation of fisheries resources and development of eco-friendly structures.
b) Support: Habitat restoration, artificial reef development, and conservation programmes.
c) Benefits: Ecological resilience, resource sustainability, and long-term environmental balance.

10.2.7. State-Level Fisheries Schemes

a) Objective: Provide region-specific support tailored to local needs.
b) Support: Construction of ponds, raceways, seed distribution, machinery support, and farmer training.
c) Benefits: Customised interventions, local employment generation, and enhanced regional production.

10.3. Conclusion:

Government schemes collectively form a comprehensive support ecosystem for fish farmers by improving access to finance, technology, infrastructure, and markets. When effectively integrated with local needs, especially in cold-water regions, these initiatives can significantly expand production, strengthen value chains, and create resilient and sustainable livelihoods.

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C. Abbreviations:

1.	**PMMSY** – *Pradhan Mantri Matsya Sampada Yojana*
2.	**Blue Revolution** – *National Programme for Fisheries and Aquaculture Development*
3.	**FIDF** – *Fisheries and Aquaculture Infrastructure Development Fund*
4.	**NFDB** – *National Fisheries Development Board*
5.	**NFDP** – *National Fisheries Development Programme*
6.	**MPEDA** – *Marine Products Export Development Authority*
7.	**NFD Fund** – *National Fisheries Development Fund*
8.	**ICAR** – *Indian Council of Agricultural Research*
9.	**CICFR** – *ICAR–Central Institute of Coldwater Fisheries Research*
10.	**DCFR** – *Directorate of Coldwater Fisheries Research* (previous name of CICFR)
11.	**RAS** – *Recirculatory Aquaculture System / Recirculating Aquaculture System*
12.	**FCR** – *Feed Conversion Ratio*
13.	**FRP** – *Fibre Reinforced Plastic*
14.	**PP** – *Polypropylene*
15.	**DO** – *Dissolved Oxygen*
16.	**pH** – *(Potential of Hydrogen)*
17.	**LPM** – *Litres Per Minute*

18.	**MT** – *Metric Tonne*
19.	**NASPAD** – *National Surveillance Programme for Aquatic Animal Diseases*
20.	**ReportFishDisease App** – Digital reporting tool under NASPAD
21.	**RBT-H** – *Rainbow Trout Heart Cell Line*
22.	**SRM-1** – *Snow Trout Muscle Cell Line*
23.	**KCC – *Kisan Credit Card***

